

DATA SCIENCE USING PYTHON - DAY WISE SYLLABUS

Duration -60 hours

S.No	Topics
1	Introduction to Data Science
2	Basics of Python Programming
3	Statistics using Python
4	Modules and packages
5	Data cleaning and manipulation
6	Descriptive statistics
7	Creating various types of plots
8	Exploring data through visualization
9	Hypothesis testing
10	Statistical analysis
11	Advanced visualization with Plotly
12	Overview of machine learning concepts
13	Supervised and unsupervised learning
14	Building and evaluating machine learning models
15	Feature engineering and selection

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16	Overview of data engineering concepts
17	Basic data analysis with NumPy
18	Data manipulation with Pandas
19	ETL (Extract, Transform, Load) processes
20	Introduction to SQL and relational databases
21	Connecting Python to databases (e.g., SQLite, MySQL)
22	Normal Distribution, Z-score,
23	Confidence Interval, Z-distribution
24	T-distribution
25	Random Forest, case study, Linear Regression
26	Hierarchical clustering and dendrogram,
27	Text mining using NLP
28	Introduction to WordCloud, Applications of wordcloud
29	Dimension Reduction, Association Rule Mining
30	Forecasting, Naive Bayes